Challenge 3: Ramsey Numbers

Definition 1 (Ramsey Number)**.** Given $t \in \mathbb{N}$, the *(diagonal) Ramsey number* $R(t)$ is the smallest $n \in \mathbb{N} \cup \{\infty\}$ such that every simple graph G on n vertices contains either a clique of size t or an independent set of size t.

By Ramsey's theorem, $R(t)$ is always an integer, which is upper bounded by 4^t [1, 2]. Erdős proved that random graphs $G(n, p)$ give a lower bound of the form $\sqrt{2}^t$ for t sufficently large [3]. An important problem that can be traced back to Ramsey's original paper [1] is to get good upper and lower bounds for the Ramsey number.

Open Problem 1.1 (Scaled Ramsey problem)**.** Let $r \in \mathbb{N}$. Provide decimal numbers d_1 and d_2 with $|d_2 - d_1| \leq (4 - \sqrt{2}) \cdot 0.9^r$ so that for all sufficiently large $t \in \mathbb{N}$

$$d_1^t \leq R(t) \leq d_2^t.$$

The scaled Ramsey problem for $r = 1$ follows by combining the above mentioned works of Ramsey and Erdős. The problem was solved for $r = 2$ by Campos, Griffiths, Morris and Sahasrabudhe [4] combined with the better bound from the follow-up paper by Gupta, Ndiaye, Norin and Wei [5].

References

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[4] M. Campos et al. "An exponential improvement for diagonal Ramsey". In: *Annals of Mathematics* 203.3 (2026), pp. 869–932.

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